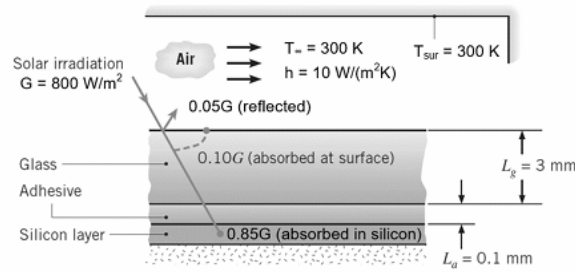


MAE 3240 Portfolio Homework Prompt

Monday, March 9, 2026 11:28 PM

(40 pt) A photovoltaic (PV) panel consists of (top to bottom) a 3-mm-thick ceria-doped glass ($k_g = 1.5$ W/m·K), a 0.1-mm-thick optical grade adhesive ($k_a = 1$ W/m·K), and a very thin layer of silicon within which solar energy is converted to



electricity. The solar-to-electricity conversion efficiency within the silicon layer η decreases with increasing silicon temperature T_{Si} , as described by the expression $\eta = a - bT_{Si}$, where $a = 0.553$ and $b = 0.001$ K⁻¹. The temperature T_{Si} is expressed in Kelvins. The backside of the silicon layer is thermally insulated. Contact thermal resistance between the adhesive and silicon layer is $R_c = 10^{-4}$ m²·K/W. There is no contact resistance between the glass and the adhesive.

In a sunny day, there is $G = 800$ W/m² solar irradiation reaching the PV panel. 5% of solar irradiation is reflected from the top surface of the glass, 10% of solar irradiation is absorbed at the top surface of the glass, and 85% of solar irradiation is transmitted to and absorbed within the silicon layer. Part of the solar irradiation absorbed in the silicon is converted to thermal energy and the remainder is converted to electricity. The PV panel is exposed to ambient air with temperature of $T_\infty = 300$ K and convective heat transfer coefficient of $h = 10$ W/m²·K. The surrounding temperature is $T_{sur} = T_\infty = 300$ K and the effective radiative heat transfer coefficient is $h_{rad} = 5$ W/m²·K. Hint: you do not need to use the Stefan-Boltzmann law to calculate the radiative heat loss.

(a) Show the thermal resistance network that can describe heat transfer across the PV panel.

Clearly label all thermal resistances, nodes for temperature, and heat fluxes entering the

temperature nodes. State key assumptions that make the thermal resistance network valid.

Calculate the value of each thermal resistance in the network.

(b) Calculate the temperatures of the top surface of the glass $T_{g,top}$ and the silicon layer T_{Si} .

(c) Calculate the solar-to-electricity conversion efficiency η and resulting electricity power flux produced by the PV panel.

MAE 3240 Portfolio Homework

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Known:

$L_g = 3 \text{ mm}$ ceria-doped glass

$L_a = 0.1 \text{ mm}$ optical grade adhesive

η = solar-to-electricity conversion efficiency

$\eta = a - bT_{si}$ silicon temperature

$a = 0.553$ $b = 0.001 \frac{1}{K}$

$G = 800 \frac{W}{m^2}$

5% solar irradiation reflected from top glass surface

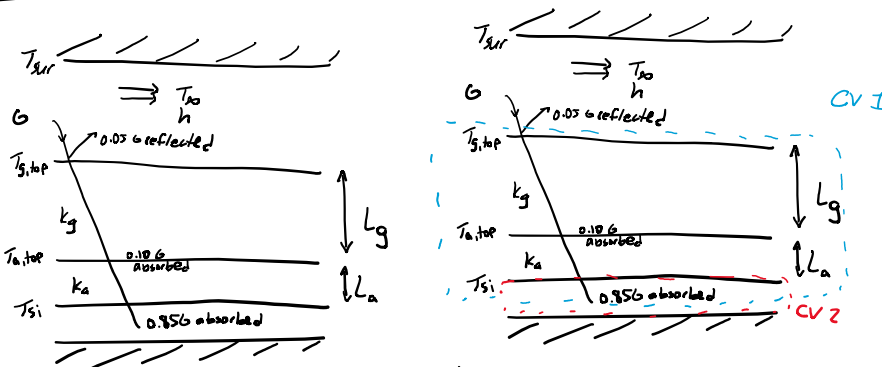
85% solar irradiation transmitted to and absorbed by silicon layer
part of absorbed solar converted to thermal energy, rest is electricity

R_c = contact thermal resistance between adhesive and silicon

Find:

- Show thermal resistance network
- state network assumptions
- find each thermal resistance
- $T_{g, \text{top}} = ?$
 $T_{si} = ?$
- $\eta = ?$
- find resulting electricity power flux produced by PV panel $P_{\text{elect}} = ?$

Schematic:



Properties:

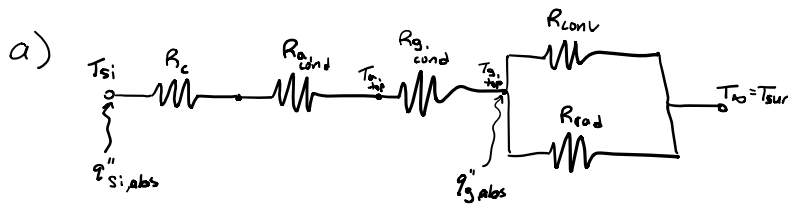
$k_g = 1.5 \frac{W}{mK}$
 $k_a = 1 \frac{W}{mK}$
 $R_c = 10^{-4} \frac{m^2K}{W}$

$T_{\infty} = 300 K$
 $T_{sur} = 300 K$
 $h_{rad} = 5 \frac{W}{m^2K}$
 $h = 10 \frac{W}{m^2K}$

Assumptions:

- don't need Stefan-Boltzmann law to calculate radiative heat loss (treat as convective)
- steady state
- 1D heat transfer

Analysis:



$$R_c'' = 10^{-4} \frac{m^2 K}{W}$$

$$R_{a,cond}'' = \frac{L_a}{k_a} = \frac{(0.1 \times 10^{-3} m)}{(1 \frac{W}{m^2 K})}$$

$$R_{a,cond}'' = 0.1 \times 10^{-3} \frac{m^2 K}{W}$$

$$R_{g,cond}'' = \frac{L_g}{k_g}$$

$$R_{g,cond}'' = \frac{L_g}{k_g} = \frac{(3 \times 10^{-3} m)}{(1.5 \frac{W}{m^2 K})}$$

$$R_{g,cond}'' = 2 \times 10^{-3} \frac{m^2 K}{W}$$

$$R_{conv} = \frac{1}{hA} \quad R_{conv}'' = \frac{1}{h} = \frac{1}{10 \frac{W}{m^2 K}}$$

$$R_{conv}'' = 0.1 \frac{m^2 K}{W}$$

$$R_{rad} \approx \frac{1}{h_{rad}A} \quad R_{rad}'' \approx \frac{1}{h_{rad}} = \frac{1}{5 \frac{W}{m^2 K}}$$

$$R_{rad}'' = 0.2 \frac{m^2 K}{W}$$

Key assumptions:

- steady state heat transfer
- 1D heat transfer
- isotropic material properties (k, h)

$$b) T_{g, \text{top}} = ?$$

$$T_{Si} = ?$$

$$q'' = \frac{\Delta T}{R''}$$

$$P''_{\text{elect}} = \eta P''_{\text{solar}} \quad P''_{\text{thermal}} = P''_{\text{solar}} - P''_{\text{elect}} \\ = P''_{\text{solar}} - \eta P''_{\text{solar}} \\ = P''_{\text{solar}} (1 - \eta)$$

CV 1:

$$\dot{E}_{\text{in}} + \dot{E}_g - \dot{E}_{\text{out}} - \dot{E}_{St} = 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$q''_{Si, \text{abs}} + q''_{g, \text{abs}} = q''_{\text{conv}} + q''_{\text{rad}}$$

$$q''_{Si, \text{abs}} = P''_{\text{thermal}} - P''_{\text{solar}} (1 - \eta) \\ = 0.856 (1 - \eta) \\ = 0.856 (1 - a + b T_{Si})$$

$$q''_{g, \text{abs}} = P''_{\text{thermal}} = 0.106$$

$$0.106 + 0.856 (1 - a + b T_{Si}) = q''_{\text{conv}} + q''_{\text{rad}}$$

$$q''_{\text{conv}} + q''_{\text{rad}} = q''_{\text{out}} = \frac{\Delta T}{R_{\text{out}}} = \frac{T_{g, \text{top}} - T_{\text{Do}}}{R_{\text{rad, conv}}}$$

$$\frac{1}{R_{\text{rad, conv}}} = \frac{1}{R_{\text{conv}}} + \frac{1}{R_{\text{rad}}} \\ = \frac{1}{(0.1 \frac{\text{m}^2 \text{K}}{\text{W}})} + \frac{1}{(0.2 \frac{\text{m}^2 \text{K}}{\text{W}})}$$

$$R_{\text{rad, conv}} = 6.67 \times 10^{-2} \frac{\text{m}^2 \text{K}}{\text{W}}$$

CV 2:

$$\dot{E}_{\text{in}} + \dot{E}_g - \dot{E}_{\text{out}} - \dot{E}_{St} = 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{E}_{\text{in}} = q''_{Si, \text{abs}} \quad \dot{E}_{\text{out}} = q''_{Si, \text{out}}$$

$$q''_{Si, \text{abs}} = 0.856 (1 - \eta) \quad q''_{Si, \text{out}} = \frac{\Delta T}{R_{eq}} = \frac{-T_{g, \text{top}} + T_{Si}}{R_c'' + R_a'' + R_{\text{cond}}''}$$

$$0.856 (1 - a + b T_{Si}) = \frac{-T_{g, \text{top}} + T_{Si}}{R_c'' + R_a'' + R_{\text{cond}}''}$$

$$R_c'' + R_a'' + R_{\text{cond}}'' = R_{eq}'' = (10^{-4} \frac{\text{m}^2 \text{K}}{\text{W}}) + (0.1 \times 10^{-3} \frac{\text{m}^2 \text{K}}{\text{W}}) + (2 \times 10^{-3} \frac{\text{m}^2 \text{K}}{\text{W}})$$

$$R_{eq}'' = 2.2 \times 10^{-3} \frac{\text{m}^2 \text{K}}{\text{W}}$$

$$0.856R''_{eq}(1-a+bT_{si}) + T_{si} = T_{g,tp}$$

$$0.106 + 0.856(1-a+bT_{si}) = \frac{T_{g,tp} - T_{A0}}{R''_{rad, conv}}$$

$$T_{g,tp} = 0.106R''_{rad, conv} + 0.856R''_{rad, conv}(1-a+bT_{si}) + T_{A0}$$

$$0.856(1-a+bT_{si}) = \frac{T_{si} - T_{g,tp}}{R''_{eq}}$$

$$0.856R''_{eq}(1-a+bT_{si}) = T_{si} - 0.106R''_{rad, conv} - 0.856R''_{rad, conv}(1-a+bT_{si}) - T_{A0}$$

$$0.106R''_{rad, conv} + 0.856R''_{rad, conv}(1-a) + 0.856R''_{eq}(1-a) + T_{A0} = T_{si} - 0.856R''_{rad, conv}bT_{si} - 0.856R''_{eq}bT_{si}$$

$$\frac{0.106R''_{rad, conv} + 0.856(1-a)(R''_{rad, conv} + R''_{eq}) + T_{A0}}{1 - 0.856R''_{rad, conv}b - 0.856R''_{eq}b} = T_{si}$$

$$T_{si} = \frac{0.106R''_{rad, conv} + 0.856(1-a)(R''_{rad, conv} + R''_{eq}) + T_{A0}}{1 - 0.856b(R''_{rad, conv} + R''_{eq})}$$

$$= \frac{0.10(800 \frac{W}{m^2})(6.67 \times 10^{-2} \frac{m^2 K}{W}) + 0.85(800 \frac{W}{m^2})(1 - 0.553)\left[(6.67 \times 10^{-2} \frac{m^2 K}{W}) + (2.2 \times 10^{-3} \frac{m^2 K}{W})\right] + (200 K)}{1 - 0.85(800 \frac{W}{m^2})(0.001 \frac{1}{K})\left[(6.67 \times 10^{-2} \frac{m^2 K}{W}) + (2.2 \times 10^{-3} \frac{m^2 K}{W})\right]}$$

$$\boxed{T_{si} = 342.3 K}$$

$$T_{g,tp} = 0.106R''_{rad, conv} + 0.856R''_{rad, conv}(1-a+bT_{si}) + T_{A0}$$

$$= 0.10(800 \frac{W}{m^2})(6.67 \times 10^{-2} \frac{m^2 K}{W}) + 0.85(800 \frac{W}{m^2})(6.67 \times 10^{-2} \frac{m^2 K}{W})[1 - 0.553 + (0.001 \frac{1}{K})(342.3 K)] + (200 K)$$

$$\boxed{T_{g,tp} = 341.1 K}$$

$$c) \eta = ?$$

$$\eta = a - bT_{Si} = (0.553) - (0.001 \frac{1}{K})(342.3 K)$$

$$\eta = 0.2107 = 21.07\%$$

$$P'_{elect} = ?$$

$$P'_{elect} = \eta P''_{solar}$$

$$= 0.856$$

$$= (0.2107)(0.85)(800 \frac{W}{m^2})$$

$$P'_{elect} = 143.3 \frac{W}{m^2}$$